

Full Length Article

CRFTrans: A recursive vision transformer reformulating the mean-field inference in conditional random field for medical image segmentation

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ABSTRACT

Transformers have recently revolutionised medical image segmentation, achieving state-of-the-art (SOTA) performance in many clinical applications. However, prevailing architectures relying on cascaded Transformer layers (e.g., 12-layer Vision Transformer) face practical limitations, leading to high computational cost and parameter redundancy. In this work, we present a new perspective that interprets Transformer operations through the paradigm of probabilistic inference. Specifically, we reveal a structural correspondence between Transformer blocks and mean-field inference in fully connected Conditional Random Fields (CRFs). Based on this insight, we propose CRFTrans, a recursive Transformer layer that reformulates the mean-field inference process using learnable attention and feed-forward operations. Unlike traditional CRFs that rely on fixed Gaussian kernels, CRFTrans leverages self-attention to model adaptive pairwise relationships in a high-dimensional feature space, enabling more expressive and data-driven contextual reasoning. We replace cascaded Transformer layers with CRFTrans in multiple state-of-the-art segmentation models and evaluate on five public datasets. Results show comparable or improved performance with significantly fewer parameters, lower memory usage, and faster training. CRFTrans provides a mathematically grounded, lightweight foundation for resource-constrained clinical deployments. The code is available in this GitHub link <https://github.com/naisops/CRFTrans/tree/main>.

1. Introduction

Medical image segmentation serves as the foundation for many clinical tasks, allowing quantitative analysis of anatomical structures in applications ranging from radiation dose calculation to intraoperative navigation systems (Shamshad et al., 2022). Driven by the rapid advancement of deep learning, convolutional neural networks (CNNs) have become the predominant approach for image segmentation. Among them, the U-Net architecture (Ronneberger et al., 2015) has achieved remarkable success due to its encoder-decoder design and multiscale skip connections, which enable effective aggregation of local spatial features. Nevertheless, the reliance on small convolutional kernels constrains CNNs to local receptive fields, hindering their capacity to capture long-range contextual dependencies across the entire image and often leading to suboptimal segmentation outcomes.

Originating in natural language processing (NLP), Transformers revolutionised sequence modelling through self-attention mechanisms (Vaswani et al., 2017). With multi-head attention (MHA) to capture global contextual relationships between the tokens (e.g., words stored in Query, Key, and Value) and the feedforward network (FFN) focus-

ing on the enhancement of local features of each token, Transformer achieves long-range dependency modeling. Vision Transformer (ViT) (Dosovitskiy et al., 2020) pioneered the Transformer's adaptation to images for the classification task through the following process: patch embedding by splitting 2D images into non-overlapping patches as image tokens; positional encoding by injecting learnable spatial coordinates; and layer stacking by processing through L identical attention-based blocks ($L = 12$ in ViT-Base). Since then, most methods have inherited this approach using Transformer layers for image segmentation tasks.

Despite their success, Transformers typically require large amounts of annotated data to achieve strong performance. This poses a significant challenge in medical image segmentation, where high-quality annotations are expensive and time-consuming to obtain. To address this limitation, many studies combined Transformers with CNNs in an encoder-decoder U-Net like structure (Chen et al., 2021b; Gong et al., 2024; Wu et al., 2022; Zhang et al., 2021), and recent studies have explored learning from AI-generated annotations or pseudo-labels to reduce the dependency on manual labelling (Song et al., 2024).

Additionally, the standard paradigm of stacking identical Transformer layers is largely empirical and lacks a clear iterative objective.

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This results in significant ‘depth-wise’ redundancy, where successive layers perform similar refinement tasks with independent, unshared parameters. We argue that this process is conceptually identical to the iterative refinement seen in probabilistic graphical models like CRFs, but the connection has remained largely unexplored.

Some methods introduced recurrence for layer simplification beyond layer stacking. For example, Universal Transformer (UT) (Dehghani et al., 2018) applies a weight-shared transition with optional adaptive halting. Sliced Recursive Transformer (SReT) (Shen et al., 2022) shares parameters across layers to improve parameter efficiency while maintaining accuracy. Task-specific Recurrent Vision Transformers (RViTs) (Gehrig & Scaramuzza, 2023) introduce stage-wise recurrence for iterative feature aggregation.

Our approach follows this line by formulating a recursive Transformer layer, but it differs in two key ways. First, rather than generic weight sharing (as in UT/SReT) that simply adopted the same Transformer layers several times, we fix Q and K across iterations and update V only, which mimics CRF-style mean-field message passing and yields sharper boundary refinement. Additionally, compared to task-specific recurrent ViTs (e.g., RViTs), our block is plug-and-play for static medical images and integrates seamlessly into encoder–decoder segmentation backbones.

Based on five Transformer-based methods, by comparing the results on five different segmentation tasks, including skin lesion segmentation, polyp segmentation, abdominal organs segmentation, cardiac segmentation, and vessel segmentation, respectively, we demonstrate the redundancy and the over-parameterisation of deep layer-stacking in the classical Transformer model for the image segmentation task. Our contributions are summarised below.

1. We provide the first formal mapping between Mean-Field (MF) inference and Transformer operations, re-interpreting Multi-Head Attention as a message-passing step with learned kernels and Feed-Forward Networks as label compatibility transforms.
2. We propose a recursive layer that replaces fixed Gaussian kernels with adaptive attention. Unlike generic recursive Transformers, CRF-Trans is uniquely designed to fix spatial queries (Q, K) while iteratively updating the latent representations (V), directly mimicking the convergence behaviour of CRF-MF inference.
3. We demonstrate that CRFTrans can replace cascaded Transformer layers in multiple state-of-the-art segmentation models, achieving comparable or improved performance with significantly fewer parameters, lower memory consumption, and faster training.

2. Related works

2.1. Transformers for image segmentation

Following the emergence of the Vision Transformer (ViT) (Dosovitskiy et al., 2020), numerous studies have explored hybrid architectures that integrate convolutional neural networks (CNNs) with Transformer modules for medical image segmentation. A common strategy involves embedding Transformer layers within the encoder to enhance global feature extraction. For instance, MedT (Qi et al., 2022) introduces gated axial-attention mechanisms to improve feature modelling within the encoder. In addition, Hatamizadeh et al. (2022) introduced UNETR, which partitions 3D medical volumes into patches and processes them through a Transformer encoder, whose multi-scale representations are subsequently decoded by a CNN-based architecture. Similarly, Wu et al. (2022) developed FAT-Net, a feature-adaptive Transformer that employs a dual-encoder design combining CNNs and Transformers for complementary feature learning. More recent works such as nnFormer (Zhou et al., 2023) further extend this idea to volumetric data using hierarchical Transformer encoders, while UNETR++ (Shaker et al., 2024) further improves multi-scale feature aggregation and representation learning.

Other works incorporated Transformer layers as a bridging mechanism between CNN encoders and decoders to capture long-range dependencies. TransUNet (Chen et al., 2021b) adopted an attention module with multiple Transformer layers to model global relationships among CNN-derived features, while TransAttUNet (Chen et al., 2021a) further enhanced this approach through the integration of Transformer self-attention and global spatial attention modules. Similarly, U-Net (Gao et al., 2021) incorporates self-attention mechanisms within intermediate feature representations to enhance long-range dependency modelling. More recent approaches such as MISSFormer (Huang et al., 2021) further explore Transformer-based feature interaction at the bottleneck, improving multi-scale representation learning.

In contrast, several models applied Transformer blocks in the decoder stage to refine contextual reasoning over features extracted by CNN encoders. For example, CTrans (Gong et al., 2024) designed a multi-resolution decoding framework where Transformer layers operate at multiple scales, maintaining hierarchical connections between encoder and decoder. Likewise, Li et al. (2021) proposed SegTran, which employs a pre-trained ResNet encoder and a squeeze-and-expansion Transformer decoder, enabling the model to learn adaptive attention matrices and diverse feature representations.

In addition, unlike conventional designs that integrate Transformer modules within encoder–decoder pipelines, TransFuse (Zhang et al., 2021) adopts a parallel architecture where CNN and Transformer branches are processed simultaneously and fused through multi-scale feature interaction.

However, compared to language models, the length of the token sequence in the image segmentation model is usually much larger. Since the computational complexity in a Transformer layer is $O(n^2)$, where n is the number of tokens, many recent studies focus on modifying the Transformer layer for more efficient learning. Efficient Transformer (Xu et al., 2021) incorporated an efficient multi-head self-attention mechanism and a novel multilayer perceptron neural network (MLP) based head to significantly lower computational complexity without sacrificing accuracy. Swin Transformer (Liu et al., 2021) introduced a shifted-window approach that limited the self-attention to local regions, drastically reducing computational complexity while still capturing global context through a hierarchical structure. Segmenter (Strudel et al., 2021) applied a point-wise linear decoder to patch embeddings, thereby leveraging global context without incurring excessive computational cost.

While reducing spatial complexity, these methods overlooked the depth-wise over-parameterisation and retained a long stack of layers. Additionally, the above spatially localised simplification sacrifices global anatomical consistency for efficiency. On the other hand, several works explore recurrence in depth by reusing the same Transformer block multiple times to improve efficiency. The Universal Transformer (UT) (Dehghani et al., 2018) is the first to explicitly formulate the Transformer as a recurrent sequence model in the depth dimension. Instead of stacking multiple independent layers, UT repeatedly applies a shared block of self-attention and feed-forward sublayers to token representations, allowing each position to evolve iteratively across recurrent steps. In computer vision, SReT (Shen et al., 2022) introduces parameter sharing across layers to enhance parameter efficiency while preserving model accuracy. Task-specific Recurrent Vision Transformers (Gehrig & Scaramuzza, 2023) further adapt the concept of recurrence for visual reasoning and event-stream processing. Instead of static feed-forward passes, RViTs perform iterative refinement over the same image or sequence features. In medical imaging, recent works adopt iterative Transformer refinements. For example, DSIT-UNet (Al Hasan et al., 2025) combined iterative transformer modules with the U-Net encoder-decoder architecture for brain-tumour segmentation, highlighting the utility of recursive attention mechanisms for medical tasks.

While current efficiency-focused Transformers like Swin (Liu et al., 2021) or Segmenter (Strudel et al., 2021) reduce spatial complexity, they still rely on layer stacking. Existing recursive models like the

Universal Transformer (UT) (Dehghani et al., 2018) or SReT (Shen et al., 2022) introduce parameter sharing to address depth-wise redundancy, yet their recurrence is often treated as a heuristic weight-sharing trick rather than a principled inference process. There remains a gap for a Transformer architecture whose recurrence is grounded in a formal refinement theory.

2.2. Conditional random field for image segmentation

CRF is a statistical modelling method that not only considers individuals but also neighbourhoods in decision-making. It is widely used in image processing tasks, such as refining the predictions of CNNs for semantic image segmentation (Chen et al., 2017; Liu et al., 2015). CRF in this context aims to increase the consistency of the labels among similar pixels, while high-dimensional features can be extracted to represent the contextual relationships of the segmentation classes (Krähenbühl & Koltun, 2011). Common CRFs contain unary potentials at the pixel level in an image and pairwise potentials that capture inter-pixel dependencies. Existing works have made efforts to explore the effective definitions of the unary and pairwise terms of CRFs. Early works focused on manually defining an effective representation to model the pairwise potentials, and the unary potential is often defined by a simple machine learning classifier (e.g. SVM) based on manually defined features. Conventional CRFs (Fulkerson et al., 2009; Gould et al., 2008) can only learn to model pairwise connections in a small neighbourhood region, resulting in excessive smoothing at object boundaries. Although a fully connected CRF is required to perform more accurate predictions around the complex object boundaries, the complex message-passing process during inference restricts its application. Krähenbühl and Koltun (2011) addressed the problem using a linear combination of Gaussian kernels to define the pairwise edge potentials, by which they proposed the mean-field (MF) algorithm, an efficient inference algorithm, for fully connected CRFs. Recent iterations, such as Gated CRFs (Obukhov et al., 2019), attempted to learn these potentials, but they lacked the high-dimensional expressive power of the multi-head attention mechanism used in our proposed framework.

In medical image segmentation, CRFs are commonly employed as post-processing techniques to refine model outputs and enhance boundary delineation. For instance, Cao et al. (2020) incorporated a CRF module into a U-Net architecture for zonal prostate segmentation on T2-weighted MRI, demonstrating improved delineation of organ boundaries. Similarly, Sun et al. (2020) proposed a hierarchical CRF framework for gastric histopathology segmentation that jointly modelled pixel-level and patch-level dependencies to better capture spatial contextual relationships within complex tissue structures. These studies highlight the potential of CRFs to improve the precision of CNN-based segmentation; however, their effectiveness remains constrained when used solely as post-processing modules, as the CRF optimisation is not jointly learned with the underlying network parameters.

Recent research has increasingly focused on achieving end-to-end integration of CRFs with deep neural networks to enable joint optimisation and contextual refinement. Zheng et al. (2015) pioneered this direction by reformulating MF inference in CRFs as a recurrent neural network (RNN), thereby allowing CRF operations to be incorporated into a CNN architecture for unified training. In medical domain, Thanh et al. (2021) integrated a CRF-RNN layer into an EfficientUNet for polyp segmentation. Subsequently, Liu et al. (2015) introduced the Deep Parsing Network (DPN), which replaced iterative MF inference with a single forward pass by designing convolutional layers capable of modelling pairwise interactions. Building on this concept, Li and Ping (2018) proposed a patch-based CRF model that utilised CNN-derived patch features to learn pairwise potentials more effectively. More recently, Chen et al. (2022) presented an end-to-end segmentation framework in which a CRF module was directly embedded within a CNN pipeline, allowing both components to be jointly optimised for improved boundary refinement and contextual consistency.

Modern end-to-end CRFs have successfully integrated MF inference into CNNs, typically by unrolling the inference steps into an RNN. However, these approaches are still limited by the use of rigid, pre-defined kernels or simple convolutional layers to model pairwise potentials. Our work bridges this gap by replacing the rigid components of CRF-MF with the flexible, learnable operations of the Transformer, creating a ‘Deep Mean-Field’ that is both mathematically sound and highly expressive.

3. Method

Both fully-connected CRF and Transformer are able to learn the long-range pairwise relationships for a given input. In the fully-connected CRF, all cliques are connected with each other, and most studies applied pre-defined functions to model the pairwise relationships. Similarly, in the Transformer layer, the pairwise relationships of all tokens are modelled by the attention layers.

In this section, we first revisit Conditional Random Fields (CRFs) and Transformer layers as two computational frameworks that share similar underlying operations. These formulations serve as the basis for establishing a structural correspondence between mean-field inference in CRFs and Transformers operations. Building on this connection, we then derive our proposed CRFTrans layer as a recursive formulation that unifies both perspectives.

3.1. Conditional random field

To establish the connection between probabilistic inference and Transformer operations, we first briefly introduce the formulation of fully connected CRFs and their mean-field inference process. This formulation will later serve as the basis for deriving a Transformer-compatible iterative refinement mechanism.

Given an image $I = \{I_1, \dots, I_N\}$ and labels $X = \{X_1, \dots, X_N\}$, the CRF models the Gibbs energy:

$$E(x) = \sum_i \Phi_u(x_i) + \sum_{i,j} \Phi_p(x_i, x_j), \quad (1)$$

where Φ_u and Φ_p denote unary and pairwise potentials, respectively.

In fully connected CRFs, the pairwise term is commonly defined by a mixture of Gaussian kernels (Krähenbühl & Koltun, 2011):

$$\Phi_p(x_i, x_j) = \mu(x_i, x_j) \sum_m \omega^{(m)} k_G^{(m)}(f_i, f_j), \quad (2)$$

where $\mu(\cdot, \cdot)$ is a compatibility function, and k_G is a Gaussian kernel over features f_i . While effective, these kernels are non-learnable and spatially rigid.

Algorithm 1 Mean-field inference process in fully-connected CRFs (Zheng et al., 2015).

```

 $Q_i(l) \leftarrow \frac{1}{Z_i} \exp(U_i(l))$  for all  $i$  ▷ Initialization
while not converged do
   $\tilde{Q}_i^{(m)}(l) \leftarrow \sum_{j \neq i} \omega^{(m)} k^{(m)}(f_i, f_j) Q_j(l)$  for all  $m$  ▷ Message Passing
   $\check{Q}_i(l) \leftarrow \sum_m \omega^{(m)} \tilde{Q}_i^{(m)}(l)$  ▷ Weighting Filter Outputs
   $\hat{Q}_i(l) \leftarrow \sum_{l' \in \mathcal{L}} \mu(l, l') \check{Q}_i(l')$  ▷ Compatibility Transform
   $\tilde{Q}_i(l) \leftarrow U_i(l) - \hat{Q}_i(l)$  ▷ Adding Unary Potentials
   $Q_i(l) \leftarrow \frac{1}{Z_i} \exp(\tilde{Q}_i(l))$  ▷ Normalising
end while

```

In a fully-connected CRF setting, due to the computational complexity, mean-field approximation is usually used for approximate inference instead of minimising the Gibbs energy directly (Krähenbühl & Koltun, 2011). Algorithm 1 shows the implementation of mean-field inference. Starting from an initial distribution derived from the unary potentials, the algorithm iteratively refines the marginal distribution through a sequence of updates. Each iteration consists of two key steps: (1) *message passing*, where pairwise interactions are aggregated across all nodes

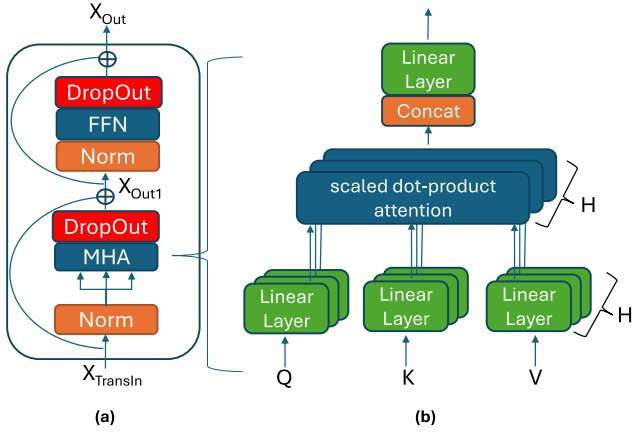


Fig. 1. The architecture of a Transformer layer. FFN indicates feed-forward networks, Norm indicates layer normalisation, MHA indicates multi-head attention layer, and Concat indicates feature concatenation. Q , K , and V indicates the query, key, and value, and H is the number of heads.

using Gaussian kernels, and (2) *compatibility transformation*, where the aggregated responses are transformed across channels (labels). The updated distribution is then combined with the unary term and normalised to produce a refined estimate. This iterative process gradually enforces spatial consistency by encouraging similar nodes to have similar label distributions, which forms the basis for our connection to Transformer operations.

3.2. Transformer

We next revisit the standard Transformer layer. In this study, the term Transformer layer refers specifically to the fundamental building block of the ViT model (Dosovitskiy et al., 2020), as illustrated in Fig. 1(a). As will be shown in Section 3.3, the key components of the Transformer, which are multi-head self-attention (MHA) and feed-forward networks (FFN), can be interpreted as counterparts to the core operations in mean-field inference.

Given input tokens $X_{Transin} \in \mathbb{R}^{N \times d}$, query (Q), key (K), and value (V) matrices are computed as:

$$Q = K = V = \text{Norm}(X_{Transin}). \quad (3)$$

where Norm indicates the normalisation layer. Then, Q , K , and V are updated by:

$$Q' = QW_Q, K' = KW_K, V' = VW_V. \quad (4)$$

where W_Q , W_K , and W_V are learnable weights. The scaled dot-product attention is defined as:

$$P = \text{Softmax}\left(\frac{Q'K'^T}{\sqrt{d_k}}\right), \quad (5)$$

$$\text{Attention}(Q', K', V') = PV'. \quad (6)$$

Multi-head attention aggregates multiple attention heads:

$$\text{MHA}(Q', K', V') = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W_O. \quad (7)$$

where W_O is the learnable linear projection.

Except for capturing the global context relationship by MHA, FFN provides a non-linear feature transformation essential for discriminating subtle patterns, enhancing the feature representation of each token:

$$\text{FFN}(X) = \text{ReLU}(XW_1 + b_1)W_2 + b_2 \quad (8)$$

where $\text{ReLU}(X) = \max(0, X)$ introduces non-linearity into the network while keeping the computation efficient.

From a computational perspective, MHA performs spatial aggregation across tokens, while FFN applies channel-wise transformation. As

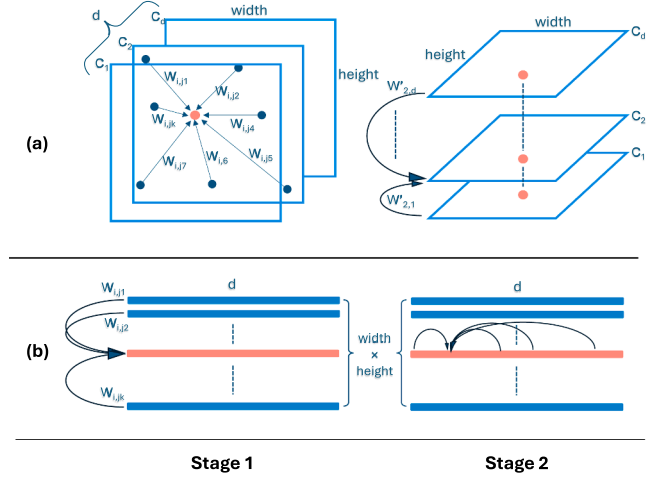


Fig. 2. Key transform steps in mean-field inference. (a) is the 2D feature map view, and (b) is the 1D feature vectors view. W and W' indicate the different weights representations in the operations from two stages. Arrows mean 'contribute to'.

shown in the next section, these operations closely correspond to the key steps in mean-field inference.

3.3. Relationship of CRF and transformer

Based on the formulations above, we observe that the Transformer's MHA performs a functionally equivalent 'message-passing' step. By comparing the iterative update of CRF with the Scaled Dot-Product Attention, we observe that the Transformer effectively replaces the rigid Gaussian affinity $k_G(f_i, f_j)$ with a learned, high-dimensional similarity $P_{i,j}$ shown in Eq. (5). This re-interpretation transforms the Transformer from a heuristic stack of layers into a principled, iterative optimiser for pairwise potentials.

Specifically, mean-field inference in CRFs and Transformer layers share a common two-stage computational structure as illustrated in Fig. 2: (1) spatial aggregation of pairwise relationships, and (2) channel-wise feature transformation. This observation reveals that Transformer operations can be interpreted as a learnable generalisation of CRF inference. Specifically, these two stages referring to spatial and channel-wise refinements in MF can be closely mapped to the MHA and FFN operations in a Transformer layer.

Stage 1: Spatial aggregation of pairwise relationships. In mean-field (MF) inference (Algorithm 1), the first stage performs spatial aggregation by combining the *Message Passing (MP)* and *Weighting Filter Outputs (WFO)* steps. Specifically, the updated probability $Q_i(l)$ is computed as a weighted sum of all other probabilities within the same class channel, as shown in stage 1 of Fig. 2:

$$Q_i(l) \leftarrow \sum_{j \neq i} W_{i,j} Q_j(l),$$

where $W_{i,j}$ corresponds to the pairwise term $k_G(f_i, f_j)$ based on spatial and intensity similarity. This operation aggregates long-range contextual information, encouraging similar pixels to have similar label distributions. In vector form (Fig. 2(b)), each feature vector is updated by aggregating all other vectors, resulting in global context integration.

A similar operation is performed in the Transformer through the scaled dot-product attention (SDPA) in MHA. Each token v'_i is updated as:

$$v'_i \leftarrow \sum_j P_{i,j} v'_j,$$

where $P_{i,j}$ is the attention weight computed from Q' and K' (Eq. (5)). Interpreting each feature vector in Fig. 2(b) as an image token, this

operation corresponds directly to the spatial aggregation in MF inference, with attention weights acting as learnable pairwise affinities.

Stage 2: Channel-wise feature transformation. The second stage of MF inference performs channel-wise transformation to refine contextual information in the label space via the *Compatibility Transform*, which mixes responses across label channels, as shown in stage 2 of Fig. 2:

$$Q_i(l) \leftarrow \sum_{l' \in \mathcal{L}} W'_{l,l'} Q_i(l'),$$

here, $W'_{l,l'}$ denotes the compatibility function $\mu(l, l')$. This operation captures inter-class relationships, refining each pixel's representation using the responses of all channels at the same spatial location.

In the view of Transformer, when each feature vector in Fig. 2(b) is treated as image token, the FFN performs a similar channel-wise transformation (see Eq. (8)). Each token is independently transformed in the embedding dimension d_{model} , which corresponds to mixing information across channels, similar to the label-space transformation in CRF.

Structural correspondence. From the above analysis, both MF inference and Transformer layers share a common two-stage structure: (1) spatial aggregation across tokens or pixels, and (2) channel-wise transformation. We utilise Fig. 2 to illustrate the structural correspondence. In CRF, the feature map shown in Fig. 2 represents a probability distribution over d classes, while in the Transformer, features lie in a d dimensional embedding space (d equals to d_{model}). Both frameworks operate on a set of elements (pixels or tokens) and model pairwise interactions among them. A fully connected CRF models pairwise relationships among all pixels, while a Transformer limits connectivity to patch tokens. Hence, the sparsity of dependencies in a Transformer is determined by the chosen patch or window size.

The key difference lies in how pairwise relationships are defined. In CRF, $W_{i,j}$ in the Stage 1 of Fig. 2 is represented by predefined Gaussian kernels $k_G(f_i, f_j)$, whereas in the Transformer, it is replaced by learnable attention weights $P_{i,j}$ in Eq. (5). Similarly, in CRF, $W'_{i,j}$ in the stage 2 of Fig. 2 is represented by the compatibility function $\mu(l_i, l_j)$, and such weights are replaced by the learnable FFN in the Transformer.

Compared to MF inference, the Transformer provides a more expressive formulation. In Stage 1, predefined Gaussian kernels restrict CRFs to modelling low-level similarities, whereas attention learns adaptive pairwise relationships in high-dimensional feature space. The multi-head mechanism further enables modelling diverse interaction patterns. In Stage 2, the linear compatibility transform in CRF is generalised to a nonlinear FFN, significantly increasing representational capacity. Overall, the Transformer layer generalises the MF process, offering superior flexibility and expressiveness for modelling long-range and cross-channel dependencies.

Notably, MF inference is inherently iterative: after each update, the refined unary term is fed back into the next iteration until convergence. In contrast, standard Transformers rely on stacking multiple layers. This observation motivates our formulation of CRFTrans, where Transformer operations are reorganised into a recursive refinement process, replacing depth with iteration.

3.4. CRFTrans

Building on the theoretical equivalence established above, we combine CRF and Transformer to propose CRFTrans—a recursive Transformer layer that reformulates MF inference as iterative hidden state refinement, shown in Fig. 3. Traditional CRF-MF iteratively updates segmentation probability maps, while CRFTrans instead refines latent feature representations, enabling feature learning during model training rather than output refinement during post-processing. Unlike conventional CRFs that rely on fixed Gaussian kernels and separate post-processing, CRFTrans replaces these components with learnable attention and feed-forward operations, enabling end-to-end training and significantly more expressive feature refinement.

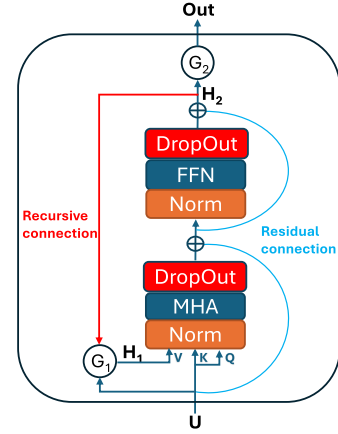


Fig. 3. The network architecture of a CRFTrans layer. U is the input unary potential. The recursive process is expressed by H_1 and H_2 , indicating the hidden states in an RNN layer, and G_1 and G_2 , indicating the gating functions.

As shown in Fig. 3, the key difference from a normal Transformer block shown in Fig. 1 is the recursive connection between the output and input, resulting in an RNN manner expressed by the following equations:

$$H_1(t) = \begin{cases} U, & t = 1, \\ H_2(t-1), & 1 < t \leq T, \end{cases} \quad (9)$$

$$H_2(t) = f_\theta(U, H_1(t)), \quad 1 \leq t \leq T, \quad (10)$$

$$Out(t) = \begin{cases} 0, & 1 \leq t < T, \\ H_2(t), & t = T. \end{cases} \quad (11)$$

where $t = 1, \dots, T$ is the iteration index, and f_θ represents the operation of one iteration in CRFTrans (θ denotes the parameters in CRFTrans). G_1 and G_2 shown in Fig. 3 are expressed as function (9) and (11). Especially, CRFTrans becomes a single Transformer layer when $T = 1$ (one forward pass without recursive process).

Algorithm 2 CRFTrans: recursive transformer layer.

```

1:  $V_i(l) \leftarrow U_i(l)$  for all  $i$  ▷ Initialisation
2: while  $t < T$  do
3:    $\tilde{V}_i(l) \leftarrow \text{Dropout} \left( MHA \begin{pmatrix} \text{Norm}(U_i(l)), \\ \text{Norm}(U_i(l)), \\ \text{Norm}(V_i(l)) \end{pmatrix} \right)$  ▷ Spatial aggregation
4:    $\check{V}_i(l) \leftarrow U_i(l) + \tilde{V}_i(l)$  ▷ Residual connection
5:    $\hat{V}_i(l) \leftarrow \text{Dropout}(\text{FFN}(\text{Norm}(\check{V}_i(l))))$  ▷ Channel transform
6:    $V_i(l) \leftarrow \check{V}_i(l) + \hat{V}_i(l)$  ▷ Residual connection
7: end while

```

We now describe the CRFTrans layer as an iterative refinement process. At each iteration, the feature representation is updated through two stages: (1) spatial aggregation using MHA (similar to MP+WFO in Algorithm 1), and (2) channel-wise transformation using FFN (similar to CompTrans in Algorithm 1). By iteratively applying these operations with shared parameters, CRFTrans performs a recursive refinement analogous to mean-field inference, while remaining fully learnable and end-to-end trainable.

The detailed procedure of CRFTrans is described in Algorithm 2. Inspired by MF, CRFTrans adopts a shared pairwise term across iterations by keeping the query and key (Q and K) representations unchanged during the recursive process, while updating only the value features. Importantly, this pairwise term is not fixed a priori, but is learned through training via the projection parameters of Q and K . As a result, the attention weights remain consistent across iterations, and tokens with higher

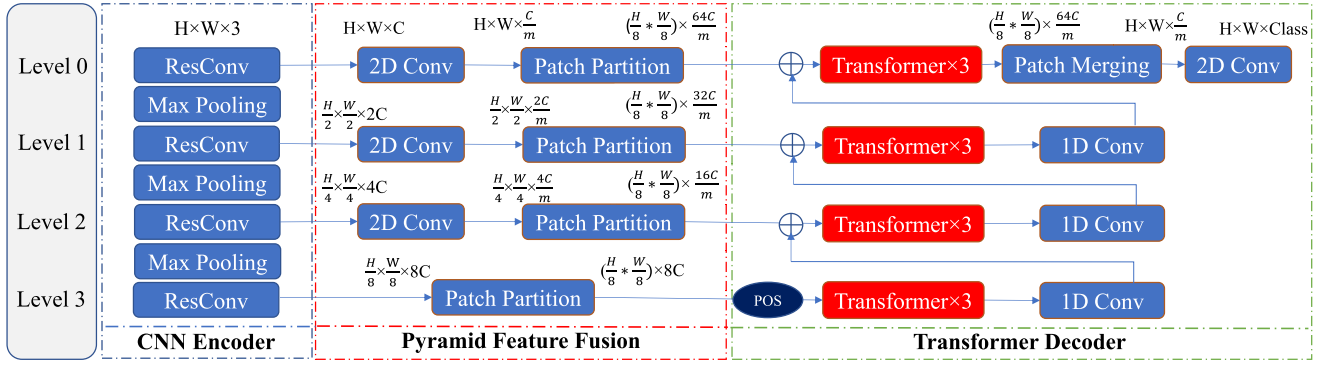


Fig. 4. The network architecture of CTranS. The red blocks are the cascaded Transformer layers that can be replaced directly by CRFTrans.

similarity are progressively brought closer in the feature space through iterative refinement.

Compared to CRF-MF, which only connects the output of each iteration to the next, CRFTrans incorporates the inner residual connections of the Transformer (as shown in Fig. 3). These residual pathways help stabilise the iterative process and mitigate gradient vanishing. In addition, CRFTrans maintains a direct connection to the initial input U , enabling each iteration to refine features while preserving the original embedding information.

Consequently, each iteration can be interpreted as combining three components: the initial embedding U , the spatial aggregation from MHA, $\tilde{V}$, and the channel-wise transformation from FFN, $\hat{V}$. The output of one iteration is therefore given by:

$$X_{out} = U + \tilde{V} + \hat{V}, \quad (12)$$

as defined in Algorithm 2.

Notation clarification. To avoid ambiguity, we distinguish variables used in CRF and Transformer formulations. In Section 3.1, $Q_i(I)$ denotes the probability distribution in mean-field inference, while in Section 3.2, Q , K , and V denote the query, key, and value matrices in the Transformer. In CRFTrans, we use U to denote the input unary features, and V to denote the iteratively refined feature representations.

In summary, CRFTrans preserves the core computational structure of mean-field inference, including iterative refinement, pairwise interaction modelling, and the two-stage update consisting of spatial aggregation and channel-wise transformation. However, several modelling assumptions are relaxed. Specifically, predefined Gaussian kernels are replaced by learnable attention mechanisms, the linear compatibility function is generalised to a nonlinear feed-forward network, and updates are performed in feature space rather than probability space. As a result, CRFTrans can be viewed as a learnable and more expressive generalisation of mean-field inference.

3.5. CRFTrans plug and play

CRFTrans is based on a single Transformer layer, it's convenient to directly replace the original cascaded Transformer layers with CRFTrans. We use CTranS (Gong et al., 2024) as an example to show the simplicity of plugging in CRFTrans. The overview of CTranS is shown in Fig. 4. It contains a CNN encoder (CE) to extract image features, a pyramid feature fusion (PFF) module to convert the CNN feature maps to image tokens, and a Transformer decoder (TD) to learn the long-range dependency of the image tokens. Cascaded Transformer layers (red blocks in Fig. 4) and CRFTrans will not change the shape of the input, and they can accept the same input size due to similar learnable layers (MHA, FFN, and Norm). Thus, we can directly replace the Transformer blocks $Transformer \times 3$ by CRFTrans. The CRFTrans can also be

easily plugged into other model backbones by replacing the Transformer blocks. We evaluate and compare them in the next section.

4. Experiments and results

In our experiments, we compare the performance of cascaded Transformer layers (Trans-stack) and CRFTrans using five SOTA Transformer-based medical segmentation models on five different medical segmentation tasks. To evaluate the effects of applying CRFTrans to different model architectures, we chose the SOTA methods that use Transformer layers in different positions. Specifically, we select: FAT-net (Wu et al., 2022), which applies Trans-stack in the encoder; TransUNet (Chen et al., 2021b), which applies Trans-stack as the bridge; CTranS (Gong et al., 2024), which applies Transformer in the decoder; and TransFuse (Zhang et al., 2021), which fuses the Transformer outputs and CNNs outputs together. Besides the hybrid models, we also compare the results with ResUNet as a baseline pure CNN-based method, and SwinUnet (Cao et al., 2021), which uses pure Transformer layers without CNN.

Moreover, Trans-stack in different models has a different number of layers according to different settings. By using l_{Tr} as the number of Trans-stack, and n_{Tr} as the number of Transformer layers in a single Trans-stack, $l_{Tr} \times n_{Tr}$ is the total number of Transformer layers in the model, which is 1×12 in FAT-Net, TransUNet, and TransFuse, 4×3 in CTranS, and 7×2 in SwinUNet. The process described in Section 3.5 was consistently applied to replace the Trans-stack by CRFTrans in all methods.

While different architectures may benefit from individually tuned hyperparameters, our goal is to evaluate the relative effectiveness of CRFTrans under consistent conditions, rather than achieving the absolute best performance for each model. Thus, to ensure a fair and controlled comparison, all models are trained using the same loss function, optimiser, learning rate, and training schedule. This unified setting allows us to isolate the impact of architectural differences without introducing variability from model-specific hyperparameter tuning. All models are trained for 150 epochs (200 epochs for FIVES dataset), which is sufficient for convergence in our experiments. We deliberately avoid specialised optimisation strategies for individual models to maintain consistency across comparisons.

4.1. Datasets and evaluation metrics

Five publicly available datasets were employed to evaluate the proposed method:

(1) **ISIC** (skin lesion segmentation): The ISIC 2018 dataset (Codella et al., 2019), released as part of the International Skin Imaging Collaboration Challenge, comprises 2594 RGB dermoscopic images of varying resolutions for binary lesion segmentation.

(2) **CVC** (polyp segmentation): The CVC-ClinicDB dataset (Bernal et al., 2015) includes 612 RGB colonoscopy frames with a resolution of

384 × 288, extracted from 31 colonoscopy sequences, and is annotated for binary polyp segmentation.

(3) **CHAOS** (abdominal organs segmentation): The CHAOS Challenge dataset (Kavur et al., 2019) provides multi-class segmentation of abdominal organs from CT and MRI scans. In this study, the T2-SPiR MRI sequence was utilised, including annotated labels for the liver, spleen, and both kidneys, across 20 subjects with 3D volumes of varying sizes.

(4) **ACDC** (cardiac segmentation): The ACDC dataset (Bernard et al., 2018), derived from clinical cardiac MRI exams at the University Hospital of Dijon, contains 3D scans of 150 patients with manual annotations for the left ventricle (LV), right ventricle (RV), and myocardium (MYO).

(5) **FIVES** (vessel Segmentation): FIVES (Jin et al., 2022) dataset consists of 800 high-resolution RGB fundus images with pixel-wise manual annotation on the vessels.

All models were implemented in a 2D framework. For volumetric (3D) data, three consecutive slices were provided as input, and the prediction was generated for the central slice using a sliding window strategy. To ensure consistency across experiments, all images were resized to 224 × 224, following the standard resolution adopted in most comparative methods. An exception was made for the fundus dataset, where images were resized to 512 × 512, as excessive downsampling would result in significant information loss in vessel segmentation tasks.

Model performance was evaluated using a 5-fold cross-validation protocol. Training was conducted on an NVIDIA GeForce GTX 2080 Ti GPU with 11 GB of memory, while the RTX A6000 (48 GB) was used for experiments involving 512 × 512 inputs. All models were optimised using the Adam optimiser with a learning rate of 0.0001 and trained for 150 epochs. During training, the checkpoint corresponding to the lowest validation loss was selected for final testing.

For evaluation, all predicted segmentation masks were rescaled to their original image dimensions prior to metric computation. Quantitative performance was assessed using the mean Dice coefficient (DC) and the average symmetric surface distance (ASSD). Statistical significance between different methods was evaluated using the Wilcoxon Signed-Rank Test (WSRT). In addition, the number of learnable parameters (NP, in millions), GPU memory consumption (GM, in gigabytes), and training time per iteration (TS, in seconds) were recorded to compare model complexity and computational efficiency.

4.2. Parameter tuning

We use CTranS with CRFTrans (explained in Section 3.5) as a plug-and-play example for parameter tuning and ablation study across multiple datasets (CVC, ISIC, CHAOS, ACDC, and FIVES). First, in parameter tuning, we explore the effects of different iterations (T) in CRFTrans. Particularly, $T = 1$ results in a normal single Transformer layer. The segmentation accuracy results are shown in Table 1.

First, Table 1 shows that $T = 3$ has an overall better performance in all datasets for the Dice measurement and some of them with statistical significance. For the ASSD measurement, $T = 3$ also outperforms other settings in most cases. For the cases that are worse than other settings, no statistical significance was found. Thus, we use $T = 3$ for all models in method comparison experiments. Note that, by comparing $T = 1$ with $T = 3$, $T = 1$ is significantly worse in most cases, which illustrates that the recursive connection benefits the single Transformer layer for more accurate segmentation. Although the single layer might lose some knowledge compared to the cascaded one, the iterative connections inspired by CRF can enhance the segmentation capability.

Furthermore, by comparing $T = 1$ with $T = 5$ and $T = 7$, it shows that the performance decreases with more iterations in some situations. Thus, more recursive times do not necessarily lead to better segmentation results, and may decrease the performance in some cases. We also observe similar trends across different model architectures, suggesting that the choice of T generalises well beyond a specific dataset or backbone. More recursive times means more refinement of the Transformer output ($H_2(t)$) based on the similarity values calculated from Q and K .

Table 1

Results of using different iteration times, expressed by T . +/− indicates better/worse average performer than $T = 3$ (**Bold ones**).

	CVC	ISIC	CHAOS	ACDC	FIVES
Dice					
$T = 1$	89.06 ^{−a}	88.33 ^{−a}	86.08 ^{−a}	89.04 [−]	84.48 [−]
$T = 3$	89.79	88.68	88.31	89.68	84.84
$T = 5$	89.27 [−]	88.43 [−]	87.63 [−]	87.44 ^{−a}	83.95 ^{−a}
$T = 7$	89.64 [−]	88.52 [−]	86.45 [−]	89.16 [−]	84.51 ^{−a}
ASSD					
$T = 1$	6.22 ^{−a}	5.06 [−]	3.98 [−]	0.51 [−]	16.74 ^{−a}
$T = 3$	5.93	5.04	3.68	0.50	14.43
$T = 5$	6.03 [−]	5.12 [−]	4.29 [−]	1.90 ^{−a}	20.24 ^{−a}
$T = 7$	5.87 ⁺	5.05 [−]	3.59 ⁺	0.49 ⁺	17.10 [−]

^a Indicates statistical significance ($p < 0.05$ measured by WSRT).

Too little refinement may result in limited knowledge learning, while too much refinement may increase the dependence on the representation of Q and K (CNN output in this case).

It is worth noting that the number of iterations in CRFTrans is not directly equivalent to the number of layers in a Trans-stack. When $T = 1$, CRFTrans reduces to a single Transformer layer, providing a direct baseline. Increasing T introduces iterative refinement with shared parameters, rather than increasing model depth with independent layers. Therefore, CRFTrans aims to replace cascaded Transformer layers through a recursive formulation, rather than matching their depth.

4.3. Ablation study

To better understand the design choices in CRFTrans, we construct several ablation variants with variant residual connections and update strategies. These variants are not proposed as new models, but are used to analyse the contribution of individual components in the CRFTrans design. We first design **CRFResidual** as an ablation variant that retains only the CRF-style residual connection while removing the inner Transformer-style residual connections, allowing us to isolate the effect of residual design as shown in Fig. 5(a). Specifically, we use a single residual connection $V_i(l) \leftarrow U_i(l) - \tilde{V}_i(l)$ after CompTrans in Algorithm 2 to replace the two residual connections (line 4 and line 6), as shown in Algorithm 3. Similarly, to evaluate the effects of seeing original features in each iteration, we design **TransResidual** as shown in Fig. 5(b), which includes a Transformer-style residual connection (no connection to the initial input during the recursive process). Specifically, we change $\tilde{V}_i(l) \leftarrow U_i(l) + \tilde{V}_i(l)$ in the Algorithm 2 to $\tilde{V}_i(l) \leftarrow V_i(l) + \tilde{V}_i(l)$. Last, we also compare CRFTrans with the general Recursive Transformer (denoted as **gRecur-Trans**, shown in Fig. 5(c)), similar to the block used in UT (Dehghani et al., 2018) and SRt (Shen et al., 2022) to explore whether fixing Q and K during the iterative refinement effectively helps. The results are shown in the Table 2.

Algorithm 3 CRFResidual.

```

1:  $V_i(l) \leftarrow U_i(l)$  for all  $i$  ▷ Initialisation
2: while  $t < T$  do
3:    $\tilde{V}_i(l) \leftarrow \text{Dropout} \left( \text{MHA} \left( \begin{matrix} \text{Norm}(U_i(l)), \\ \text{Norm}(U_i(l)), \\ \text{Norm}(V_i(l)) \end{matrix} \right) \right)$  ▷ MP + WFO
4:    $\hat{V}_i(l) \leftarrow \text{Dropout}(\text{FFN}(\text{Norm}(\tilde{V}_i(l))))$  ▷ CompTrans
5:    $V_i(l) \leftarrow U_i(l) - \hat{V}_i(l)$  ▷ Residual connect
6: end while

```

As shown in Table 2, in most cases, CRFTrans has better mean performance than gRecur-Trans, with more statistical significance in the

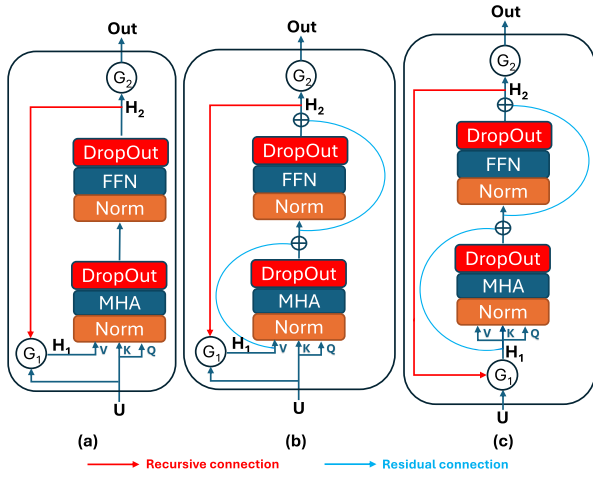


Fig. 5. The network architecture of CRFResidual (a), TransResidual (b), and gRecur-Trans (c).

Table 2

Results of ablation study. +/− indicates better/worse average performer than original setting (**Bold ones**).

	CVC	ISIC	CHAOS	ACDC	FIVES
Dice					
CRFTrans	89.79	88.68	88.31	89.68	84.84
CRFResidual	88.85 ^{−a}	88.44 ^{−a}	86.71 ^{−a}	88.87 [−]	84.21 ^{−a}
TransResidual	89.27 [−]	88.66 [−]	87.77 [−]	89.58 [−]	85.03 ⁺
gRecur-Trans	89.42 [−]	88.86 ⁺	88.00 [−]	89.35 [−]	84.02 ^{−a}
ASSD					
CRFTrans	5.93	5.04	3.68	0.50	14.43
CRFResidual	7.17 ^{−a}	4.99 ⁺	4.45 [−]	0.61 [−]	17.01 [−]
TransResidual	6.40 [−]	4.99 ⁺	3.67 ⁺	0.51 [−]	16.34 ^{−a}
gRecur-Trans	8.38 ^{−a}	5.16 ^{−a}	4.37 [−]	0.45 ⁺	24.88 ^{−a}

^a Indicates statistical significance ($p < 0.05$ measured by WSRT).

ASSD measure, demonstrating the advantage of CRFTrans in sharper boundary detection compared with generic weight sharing. Additionally, CRFResidual performs significantly worse than CRFTrans, but the results of TransResidual have no statistically significant difference compared with CRFTrans. It demonstrates that the inner skip connections on $MP + WFO$ and CompTrans significantly benefit the CRFResidual that only considers the CRF-style residual connection. On the other hand, whether to see the original input in each recursive process has no significant effect in most cases. However, we still use CRFTrans that combines both CRF-style and Transformer-style residual connections rather than TransResidual in the latter experiments, since CRFTrans has better average results, and in situations requiring more recursive times, the training process can be more stable by seeing the initial input in each iteration.

4.4. State-of-the-art model comparison

After determining the model settings, we compare the performance of CRFTrans with Trans-stack. For a fair comparison, we retain the original Trans-stack configurations in each model and replace them with CRFTrans, without modifying the overall architecture. This setup evaluates whether CRFTrans can serve as a drop-in replacement for cascaded Transformer layers. Additionally, in CRFTrans, we use the same parameter settings as a single layer in Trans-stack. Furthermore, although pretrained Transformer blocks are used in the original FAT-Net, TransUNet, TransFuse, and SwinUNet as a default setting, we train all layers from scratch for a fair comparison with CRFTrans. Our evaluation covers encoder-, bridge-, decoder-, and fusion-based Transformer designs,

Table 3

Dice of segmentation of different models. +/− indicates better or worse performance for the results that have a statistically significant difference.

	CVC	ISIC	CHAOS	ACDC	FIVES
ResUNet					
	76.07	76.37	84.95	88.35	86.59
FAT-Net					
Trans-stack	88.07 ^a	87.17 ^a	87.03 ^a	89.84 ^a	87.52
CRFTrans	89.20	87.29	86.35	89.10	88.53 ⁺
TransUNet					
Trans-stack	86.19 ^a	87.99 ^a	85.89 ^a	88.69 ^a	87.91
CRFTrans	85.65	88.08	86.17	88.00	89.64 ⁺
SwinUNet					
Trans-stack	75.99	87.15	76.53 ^a	82.96	86.14
CRFTrans	73.73 [−]	85.69 [−]	74.37	80.97 [−]	85.16 [−]
CTrans					
Trans-stack	90.01 ^a	89.07 ^a	88.51 ^a	89.68 ^a	84.98 ^a
CRFTrans	89.79	88.68	88.31	89.68	84.84
TransFuse					
Trans-stack	84.76	87.92	83.88 ^a	87.94 ^a	87.34
CRFTrans	84.11 [−]	87.60 [−]	85.01	88.34	86.61 [−]

^a Indicates no statistical significance compared to CRFTrans ($p > 0.05$ measured by WSRT).

providing a comprehensive assessment of CRFTrans across different architectural paradigms.

Tables 3 and 4 present the Dice and ASSD metrics of the segmentation results. Overall, for the four CNN-Transformer hybrid models (FAT-Net, TransUNet, CTrans, and TransFuse), CRFTrans achieves performance comparable to Trans-stack in most cases, with no statistically significant differences observed in the majority of settings. Although a slight performance drop is observed in a few cases for TransFuse, the degradation is marginal. These results demonstrate that CRFTrans can effectively replace cascaded Transformer layers across different hybrid architectures without sacrificing segmentation accuracy, regardless of whether the Transformer is used in the encoder, bridge, decoder, or fusion stage. Furthermore, CRFTrans shows notable improvements in vessel segmentation (FIVES) in several cases. This suggests that the recursive refinement mechanism in CRFTrans is particularly beneficial for challenging tasks involving small and thin structures, where iterative similarity-driven updates can better capture fine-grained details compared to conventional layer stacking.

However, as shown in Tables 3 and 4, CRFTrans exhibits significant reduced performance when applied to SwinUNet, indicating its limited effectiveness in pure Transformer architectures. This behaviour can be explained by the reliance of CRFTrans on the quality of the pairwise similarity, which is determined by the representations of Q and K . Since the recursive refinement process is guided by the similarity matrix derived from these representations, the effectiveness of CRFTrans depends on a reliable unary feature embedding. In hybrid models such as TransUNet, CTrans, and TransFuse, this requirement is satisfied by the presence of CNN-based feature extraction or fusion mechanisms, which provide strong and stable representations. In particular, TransFuse adopts a parallel fusion strategy that combines CNN and Transformer features, offering complementary feature sources that help stabilise the refinement process. Similarly, in FAT-Net, although CRFTrans is applied in the encoder, the dual-encoder design mitigates excessive dependence on a single representation source.

In contrast, in pure Transformer architectures such as SwinUNet, the feature representations are progressively generated by stacked Transformer layers. When these layers are replaced by CRFTrans, the

Table 4

ASSD of segmentation of different models. +/− indicates better or worse performance for the results that have a statistically significant difference.

	CVC	ISIC	CHAOS	ACDC	FIVES
ResUNet					
	18.99	12.13	1.65	0.53	16.05
FAT-Net					
Trans-stack	7.24 ^a	5.80 ^a	3.30 ^a	0.31 ^a	16.30
CRFTrans	7.46	5.87	3.29	0.38	17.94 [−]
TransUNet					
Trans-stack	11.43 ^a	5.53 ^a	3.23 ^a	0.39 ^a	14.12
CRFTrans	10.61	5.74	1.25	0.40	15.83 [−]
SwinUNet					
Trans-stack	17.24 ^a	5.80	2.11 ^a	1.63	9.94
CRFTrans	14.87	6.46 [−]	4.13	0.85 ⁺	7.82 ⁺
CTranS					
Trans-stack	5.82 ^a	4.72 ^a	3.72 ^a	0.47 ^a	14.88
CRFTrans	5.93	5.04	3.68	0.50	14.43 ⁺
TransFuse					
Trans-stack	16.57 ^a	6.02 ^a	3.87 ^a	0.38 ^a	14.98 ^a
CRFTrans	23.40	6.02	3.43	0.41	18.07

^a Indicates no statistical significance compared to CRFTrans ($p > 0.05$ measured by WSRT).

recursive refinement becomes dependent on its own intermediate representations, which may be insufficiently expressive in early stages. As a result, the model cannot effectively learn hierarchical features, leading to degraded performance compared to the original Trans-stack design. This observation highlights that CRFTrans is particularly suitable for CNN-Transformer hybrid architectures, where reliable feature representations are available to guide the recursive refinement process.

Moreover, by comparing the baseline pure CNN model and pure Transformer model with the hybrid models, the hybrid models perform better in most cases, demonstrating the benefit from the combination, which fully utilises the advantage of CNN to learn local high-level features and the advantage of Transformer to capture long-term dependency.

Furthermore, the models' complexity, memory consumption and training efficiency measured by NP, GM, and TS are listed in Table 5. The results show that CRFTrans consistently reduces model complexity, achieving significantly fewer parameters (NP) and lower memory consumption (GM), across all architectures except for SwinUNet, where CRFTrans consumes slightly more memory. CRFTrans reduces the total number of Transformer layers from $l_{Tr} \times n_{Tr}$ to l_{Tr} by replacing stacked layers with a recursive formulation. The computational cost is instead controlled by the number of iterations T . In our experimental setting, the effective operation times per CRFTrans are 3, which is substantially lower than those used in FAT-Net, TransUNet, and TransFuse (12). For CTranS and SwinUNet, the operation times of Trans-stack (3 and 2, respectively) are comparable to CRFTrans, resulting in less pronounced efficiency gains, and even slightly more memory consumption for SwinUNet. Consequently, the reduction in memory consumption and training time is more significant in models with deeper Transformer stacks, such as FAT-Net, TransUNet, and TransFuse.

In practice, when replacing Trans-stack with CRFTrans, we recommend starting with a small number of iterations (e.g., $T = 1$) and gradually increasing T until the performance matches that of the original Trans-stack. This strategy enables maximum model simplification while maintaining accuracy, as larger values of T introduce additional computational cost without consistently improving performance.

Table 5

Results of model complexity comparison.

	NP	GM	TS
ResUNet			
	13.04	1.42	0.03
FAT-Net			
Trans-stack	26.98	1.03	0.14
CRFTrans	23.72	0.90	0.05
TransUNet			
Trans-stack	105.28	2.46	0.13
CRFTrans	27.31	1.11	0.07
SwinUNet			
Trans-stack	27.17	0.90	0.09
CRFTrans	15.41	0.91	0.08
CTranS			
Trans-stack	21.48	2.14	0.09
CRFTrans	11.75	2.07	0.06
TransFuse			
Trans-stack	33.35	0.81	0.13
CRFTrans	13.83	0.46	0.06

Notably, although CRFTrans adopts a recursive formulation with shared parameters, several design choices ensure stable training. Each iteration includes residual connections and layer normalisation, which help maintain gradient flow and stabilise feature distributions, similar to standard Transformer architectures. In addition, the number of iterations is small in practice (e.g., $T = 3$), avoiding long recursive chains. Empirically, we observe stable convergence across all experiments without requiring additional stabilisation techniques.

5. Conclusion

In this study, we present a novel perspective on Transformer architectures by establishing a connection between Transformers and mean-field inference in CRFs. Based on this insight, we propose CRFTrans, a recursive Transformer block that reformulates cascaded Transformer layers as an iterative refinement process. Extensive experiments across five Transformer-based segmentation models and five medical imaging datasets demonstrate that CRFTrans can significantly reduce model parameters and memory consumption while maintaining competitive segmentation performance for CNN-Transformer hybrid models. Notably, for challenging tasks such as vessel segmentation, CRFTrans generally achieves improved accuracy, suggesting that recursive refinement is particularly effective for capturing fine-grained structures.

Although our experiments are conducted in a 2D framework for fair comparison with existing methods, CRFTrans is not restricted to 2D inputs. Similar to Transformers, it operates on tokenised feature representations and can be naturally extended to 3D by applying the same iterative refinement mechanism to volumetric tokens. From a CRF perspective, the formulation also generalises to arbitrary graph structures, including 3D grids. In practice, the main challenge in 3D settings lies in the increased computational cost due to the larger number of tokens, which can be mitigated using efficient attention mechanisms.

Overall, this work shows that Transformer depth can be replaced by iterative inference, providing a more efficient and theoretically grounded alternative to conventional layer stacking. This perspective opens up new directions for designing lightweight and expressive architectures for medical image segmentation.

However, as the main limitation of CRFTrans, the dependence on the quality of unary terms hinders its application in pure Transformer-based models. In future work, besides segmentation tasks, we will

explore wider applications of CRFTrans (e.g. image classification and detection tasks) to research on the generalisability of replacing Trans-stack with CRFTrans as backbone models.

CRedit authorship contribution statement

Zhendi Gong: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization; **Guoping Qiu:** Writing – review & editing, Supervision; **Xin Chen:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xin Chen reports financial support was provided by University of Nottingham. Guoping Qiu reports financial support was provided by University of Nottingham Ningbo China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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